1 · Dynamics of expectation values

- **Generalized Ehrenfest:** $\frac{d}{dt}\langle Q\rangle = \frac{i}{\hbar}\langle[\hat H,\hat Q]\rangle + \langle \partial_t \hat Q\rangle$. Commutes with $\hat H$ (no explicit t) ⇒ **conserved**.
- $[\hat A\hat B,\hat C] = \hat A[\hat B,\hat C] + [\hat A,\hat C]\hat B \cdot [f(\hat x),\hat p] = i\hbar f' \cdot [\hat x,\hat p^2] = 2i\hbar\hat p \cdot [\hat H,\hat x] = -\frac{i\hbar}{m}\hat p \cdot [\hat H,\hat p] = i\hbar V'$.
- $Q=1$: norm conserved; $\hat H$: energy; $\hat x$: $\frac{d\langle x\rangle}{dt} = \frac{\langle p\rangle}{m}$; $\hat p$: $\frac{d\langle p\rangle}{dt} = -\langle V'\rangle$ (Newton only if $V \le$ quadratic).
- Virial: $\frac{d}{dt}\langle xp\rangle = 2\langle T\rangle - \langle xV'\rangle = 0$ in stationary states. $V\propto x^2$: $\langle T\rangle = \langle V\rangle$; $V\propto -1/r$: $\langle T\rangle = -E$, $\langle V\rangle = 2E$.
- Stationary state: every $\langle Q\rangle$ constant (no explicit t). Parity: even V ⇒ $\hat\Pi$ conserved.

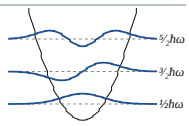
2 · Uncertainty, energy and time

- $\sigma_A\sigma_B \ge \left|\frac{1}{2i}\langle[\hat A,\hat B]\rangle\right| \cdot \sigma_H\sigma_Q \ge \frac{\hbar}{2}\left|\frac{d\langle Q\rangle}{dt}\right|$ ⇒ $\Delta E\Delta t \ge \frac{\hbar}{2}$, $\Delta t = \frac{\sigma_Q}{|d\langle Q\rangle/dt|}$.
- $\sigma_x\sigma_H \ge \frac{\hbar}{2m}|\langle p\rangle|$: empty for stationary states (both sides 0). Lifetime τ ⇒ linewidth $\sim \hbar/\tau$. Fast change needs big σ_H.
- Equality iff $(\hat B - \langle B\rangle)\psi = i\lambda(\hat A - \langle A\rangle)\psi$, λ real (Schwarz equality + $\mathrm{Re}\,z = 0$). x,p: Gaussian, $\sigma_x^2 = \hbar/2\lambda$. Free Gaussian leaves the bound as $\mathrm{Re}\,z = \frac{1}{2}\langle xp+px\rangle - \langle x\rangle\langle p\rangle$ grows. ISW can't reach $\hbar/2$ (no Gaussian vanishes at walls).

3 · Oscillator: algebra

- $\hat a_\pm = \frac{1}{\sqrt{2\hbar m\omega}}(\mp i\hat p + m\omega\hat x)$, $[\hat a_-,\hat a_+]=1$, $\hat H = \hbar\omega(\hat a_+\hat a_- + \frac12)$, $\hat a_\pm^\dagger = \hat a_\mp$.
- $\hat H\hat a_\pm\psi = (E\pm\hbar\omega)\hat a_\pm\psi$; ladder ends: $\hat a_-\psi_0 = 0$ ⇒ $E_n = \hbar\omega(n+\frac12)$, $n=0,1,\dots$; evenly spaced.
- $\hat a_+|n\rangle = \sqrt{n+1}\,|n+1\rangle$, $\hat a_-|n\rangle = \sqrt{n}\,|n-1\rangle$, $|n\rangle = \frac{(\hat a_+)^n}{\sqrt{n!}}|0\rangle$, $\hat N = \hat a_+\hat a_-$.
- $\hat x = \sqrt{\frac{\hbar}{2m\omega}}(\hat a_+ + \hat a_-)$, $\hat p = i\sqrt{\frac{\hbar m\omega}{2}}(\hat a_+ - \hat a_-)$. **x, p: n→n±1; x², p²: n→n, n±2.**
- $|n\rangle$: $\langle x^2\rangle = (n+\frac12)\frac{\hbar}{m\omega}$, $\langle p^2\rangle = (n+\frac12)\hbar m\omega$, $\sigma_x\sigma_p = (n+\frac12)\hbar$, $\langle T\rangle = \langle V\rangle = \frac{E_n}{2}$.
- $\langle n+1|\hat x|n\rangle = \sqrt{\frac{(n+1)\hbar}{2m\omega}}$; $\langle n+2|\hat x^2|n\rangle = \frac{\hbar}{2m\omega}\sqrt{(n+1)(n+2)}$.

4 · Oscillator: wavefunctions and motion



- $\psi_0 = (\frac{m\omega}{\pi\hbar})^{1/4}e^{-m\omega x^2/2\hbar}$; $\psi_n \propto H_n(\xi)e^{-\xi^2/2}$, $\xi = \sqrt{\frac{m\omega}{\hbar}}x$, norm $(\frac{m\omega}{\pi\hbar})^{1/4}(2^n n!)^{-1/2}$. $H_1 = 2\xi$, $H_2 = 4\xi^2 - 2$, $H_3 = 8\xi^3 - 12\xi$.
- n nodes, parity $(-1)^n$, tails beyond turning point $x_n = \sqrt{2n+1}\,w_x$, $w_x = \sqrt{\hbar/m\omega}$. Large n → $P_{cl} = \frac{1}{\pi\sqrt{x_0^2 - x^2}}$ (peaks at the turning points).
- Half oscillator (wall at 0): odd n only, $E = \frac32, \frac72, \dots \hbar\omega$, $\psi = \sqrt2\psi_{odd}$. $V + qEx$: complete the square, shift by $-qE/m\omega^2$, E_n lowered by $\frac{q^2E^2}{2m\omega^2}$.
- $c_0|0\rangle + c_1|1\rangle$: $\langle x\rangle = \sqrt{\frac{2\hbar}{m\omega}}|c_0c_1|\cos(\omega t - \theta)$; phases: $\langle H\rangle$ gives $|c_n|$, $\langle x\rangle$ the relative phase, sign $\langle p\rangle$ its sign.
- Phase space $(\langle x\rangle, \langle p\rangle/m\omega)$: clockwise circle, period $2\pi/\omega$. $|n\rangle$: a point at 0. $|0\rangle \pm |2\rangle$: $\langle x\rangle = 0$, width breathes at 2ω.
- Classical iff $n \gg 1$ and amplitude $x_0 \gg w_x$ ($x_0/w_x = \sqrt{2n+1}$).

5 · Coherent states

- $\hat a_-|\alpha\rangle = \alpha|\alpha\rangle$, $|\alpha\rangle = e^{-|\alpha|^2/2}\sum\frac{\alpha^n}{\sqrt{n!}}|n\rangle$, $P(n) = e^{-|\alpha|^2}\frac{|\alpha|^{2n}}{n!}$ (Poisson).
- $\langle n\rangle = |\alpha|^2$, $\sigma_n = |\alpha|$, $\sigma_H/\langle H\rangle \to 1/|\alpha|$. $\alpha(t) = \alpha e^{-i\omega t}$: stays coherent.
- $\langle x\rangle = \sqrt{\frac{2\hbar}{m\omega}}\mathrm{Re}\,\alpha$, $\langle p\rangle = \sqrt{2\hbar m\omega}\,\mathrm{Im}\,\alpha$; $\sigma_x = \sqrt{\frac{\hbar}{2m\omega}}$, $\sigma_p = \sqrt{\frac{\hbar m\omega}{2}}$ always: minimum uncertainty, ground-state blob riding the classical orbit. $|0\rangle = |\alpha = 0\rangle$.
- vs $|N\rangle$ ($|\alpha|^2 = N$): same $\langle H\rangle$; $|N\rangle$ has $\sigma_H = 0$ and doesn't move. Motion needs adjacent n with fixed relative phases; price: $\sigma_H = \hbar\omega|\alpha|$.

Exam moves: see this → do this

- **Superposition of $|n\rangle$** → write $\hat x|\psi\rangle$ with ladders; only $n, n\pm1$ survive; frequencies = energy differences.
- **Known system, two states** → it's a qubit: σ_z = which level, σ_x = which side.
- **Function of θ, φ** → Cartesian shapes: $x \pm iy$, z, $(x+iy)^2$ → read off ℓ, m; $e^{im\phi}$ alone gives L_z.
- **Measure L_x in ℓ=1** → 3×3 matrix, eigenvectors $\frac12(1,\pm\sqrt2,1)$, $\frac{1}{\sqrt2}(1,0,-1)$.
- **Hermitian?** → by parts; say which boundary condition kills the boundary term.
- **Central V** → $u = rR$, sketch V_{eff}, mark allowed regions, count nodes.
- **T/F** → verdict + the principle (conservation, $u(0)=0$, bound ⇔ $E < V(\infty)$, collapse, compatibility). Verdict alone = 1/5.

6 · Three dimensions

- $\hat{\vec p} = -i\hbar\nabla$; $\nabla^2 = \frac{1}{r^2}\partial_r(r^2\partial_r) - \frac{\hat L^2}{\hbar^2r^2}$; $d^3r = r^2\sin\theta\,dr\,d\theta\,d\phi$, $d\Omega = \sin\theta\,d\theta\,d\phi$.
- Central $V(r)$: $\psi = R(r)Y_\ell^m$, separation constant $\ell(\ell+1)$; $\int|R|^2r^2dr = 1$, $\int|Y|^2d\Omega = 1$.
- Cube: $E \propto n_x^2 + n_y^2 + n_z^2$. Isotropic SHO: $E = \hbar\omega(N + \frac32)$, deg. $\frac{(N+1)(N+2)}{2}$, $\ell = N, N-2, \dots$ (N=2: $\ell = 0 + \ell = 2$: accidental).
- Classically $\vec L$ fixed ⇒ plane (3D→2D), $|L|$ fixed ⇒ radial 1D. QM keeps only the second ($\hat L^2$ sharp, direction not).

7 · Angular momentum algebra

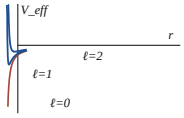
- $\hat L_z = \hat x\hat p_y - \hat y\hat p_x$ (cyclic). $[\hat L_x,\hat L_y] = i\hbar\hat L_z$, $[\hat L^2,\hat L_i] = 0$, $[\hat L_z,\hat x] = i\hbar\hat y$, $[\hat L_z,\hat y] = -i\hbar\hat x$, $[\hat L_z,\hat z] = 0$ (same with $\hat p$). $[\hat L_z, r^2] = [\hat L_z, p^2] = 0$.
- Only one component sharp (unless $\ell = 0$). Compatible set $\{\hat H, \hat L^2, \hat L_z\}$ for central V; $\frac{d}{dt}\langle\vec L\rangle = \langle\vec r\times(-\nabla V)\rangle$.
- $\hat L_\pm = \hat L_x \pm i\hat L_y$, $\hat L_\pm^\dagger = \hat L_\mp$, $[\hat L_z,\hat L_\pm] = \pm\hbar\hat L_\pm$, $\hat L_\mp\hat L_\pm = \hat L^2 - \hat L_z^2 \mp \hbar\hat L_z$.
- $\hat L^2 = \hbar^2\ell(\ell+1)$, $L_z = \hbar m$, $m = -\ell..\ell$; $\hat L_\pm Y_\ell^m = \hbar\sqrt{\ell(\ell+1) - m(m\pm1)}\,Y_\ell^{m\pm1}$; $\hat L_+Y_\ell^\ell = 0$.
- Vector model: $|L| = \hbar\sqrt{\ell(\ell+1)} > \hbar\ell$; cone angle $\cos\theta = m/\sqrt{\ell(\ell+1)}$; $\langle L_x^2\rangle = \langle L_y^2\rangle = \frac{\hbar^2}{2}[\ell(\ell+1) - m^2]$ in Y_ℓ^m.
- ℓ bounds m; m only gives $\ell \ge |m|$.

8 · Spherical harmonics, measuring L

- $\hat L_z = -i\hbar\partial_\phi$ ⇒ $e^{im\phi}$, $m\in\mathbb{Z}$ (single-valued). $\hat L_\pm = \pm\hbar e^{\pm i\phi}(\partial_\theta \pm i\cot\theta\,\partial_\phi)$ ⇒ $Y_\ell^\ell \propto \sin^\ell\theta\,e^{i\ell\phi}$.
- $Y_0^0 = \frac{1}{\sqrt{4\pi}}$, $Y_1^0 = \sqrt{\frac{3}{4\pi}}\cos\theta$, $Y_1^{\pm1} = \mp\sqrt{\frac{3}{8\pi}}\sin\theta e^{\pm i\phi}$, $Y_2^0 = \sqrt{\frac{5}{16\pi}}(3\cos^2\theta - 1)$, $Y_2^{\pm1} = \mp\sqrt{\frac{15}{8\pi}}\sin\theta\cos\theta e^{\pm i\phi}$, $Y_2^{\pm2} = \sqrt{\frac{15}{32\pi}}\sin^2\theta e^{\pm2i\phi}$. Parity $(-1)^\ell$.
- $\ell = 1$ basis $(1,0,-1)$: $L_x = \frac{\hbar}{\sqrt2}\begin{pmatrix}0&1&0\\1&0&1\\0&1&0\end{pmatrix}$, $L_y = \frac{\hbar}{\sqrt2}\begin{pmatrix}0&-i&0\\i&0&-i\\0&i&0\end{pmatrix}$. L_x: $\pm\hbar$: $\frac12(1,\pm\sqrt2,1)$; 0: $\frac{1}{\sqrt2}(1,0,-1)$. From $Y_1^{\pm1}$: $P(L_x = \hbar,0,-\hbar) = \frac14,\frac12,\frac14$; from Y_1^0: $\frac12, 0, \frac12$.
- Successive: measuring L_x erases L_z; L^2 survives. Probabilities need the full state, not just ℓ.
- Rotor $\hat H = \frac{\hat L^2}{2I}$: $E_\ell = \frac{\hbar^2\ell(\ell+1)}{2I}$, gaps $\frac{\hbar^2}{I}(\ell+1)$, deg. $2\ell+1$. $I_z \ne I$: $E = \frac{\hbar^2}{2I}[\ell(\ell+1) - \frac{\epsilon}{1+\epsilon}m^2]$, $\pm m$ stay degenerate; $-\gamma B\hat L_z$ splits them.

9 · Radial equation, V_{eff}, box

- $-\frac{\hbar^2}{2m}u'' + V_{\text{eff}}u = Eu$, $V_{\text{eff}} = V + \frac{\hbar^2\ell(\ell+1)}{2mr^2}$; $u(0)=0$ (ℓ=0: $1/r$ would give a δ^3); $|u|^2dr$ = radial probability.



- Near 0: $u \propto r^{\ell+1}$, $R \propto r^\ell$: only s-states have $\psi(0)\ne0$. Bound: $u\to0$, discrete; $E > V(\infty)$: oscillates, continuous. $+k/r$: no bound states.
- Spherical box b: $E = \frac{\hbar^2\beta^2}{2mb^2}$; ℓ=0: $u = \sin kr$, $\beta = N\pi$. Lowest ℓ ground state (centrifugal term > 0). Order 1s 1p 1d 2s 1f; magic 2, 8, 20.

	N=1	2	3
ℓ=0	3.142	6.283	9.425
1	4.493	7.725	10.90
2	5.763	9.095	
3	6.988	10.42	
4	8.183		

10 · Hydrogen

- $V = -\frac{k}{r}$, $a = \frac{\hbar^2}{mk} = 0.0529$ nm, $E_n = -\frac{k}{2an^2} = -\frac{13.6\,\text{eV}}{n^2}$, $n = N+\ell$, $\ell \le n-1$; deg. n^2 (2ℓ+1 rotational, rest accidental: hidden $1/r$ symmetry).
- $\psi_{100} = \frac{e^{-r/a}}{\sqrt{\pi a^3}}$; $P(r) = \frac{4r^2}{a^3}e^{-2r/a}$, peak a, $\langle r\rangle = \frac32 a$, $\langle\frac1r\rangle = \frac1a$; $P(r<b) = 1 - e^{-2b/a}(1 + \frac{2b}{a} + \frac{2b^2}{a^2})$, $P(r<a) = 0.323$.
- $R_{20} = \frac{1}{\sqrt{2a^{3/2}}}(1 - \frac{r}{2a})e^{-r/2a}$, $R_{21} = \frac{1}{\sqrt{24a^{3/2}}}\frac{r}{a}e^{-r/2a}$. $\langle r\rangle = \frac{a}{2}[3n^2 - \ell(\ell+1)]$, $\langle\frac1r\rangle = \frac{1}{n^2a}$.
- V_{eff} min at $r = \ell(\ell+1)a$, depth $E_1/\ell(\ell+1)$. $\lambda = \frac{1240\text{ eV nm}}{\Delta E}$: Lyman α 122 nm; Balmer 656, 486, 434 nm.
- Extra $-\frac{\hbar^2\alpha^2}{2mr^2}$ (alkali core): $\ell \to \lambda_\ell$, $E = \frac{E_1}{(n-\delta_\ell)^2}$, $\delta_\ell \approx \frac{\alpha^2}{2\ell+1}$: s lowest (penetrates).

Traps

- $\hat a_-|0\rangle = 0$, not $|-1\rangle$; n starts at 0. $\hat a_\pm$ not Hermitian.
- $\hat x|n\rangle$ has **two** terms; $\langle x^2\rangle$ needs $\hat a_-\hat a_+$ and $\hat a_+\hat a_-$ (they differ by 1).
- $|\vec L|^2 = \hbar^2\ell(\ell+1)$, not $\hbar^2\ell^2$. $Y_1^{\pm1}$ carries $\mp$.
- $d\Omega$ has $\sin\theta$; $r^2|R|^2$ (not $|R|^2$) is the radial density.
- Energy can't leak or be "lost" with time-independent $\hat H$; bound needs $E < V(\infty)$.

Constants

$\hbar = 6.582\times10^{-16}$ eV·s $= 1.055\times10^{-34}$ J·s · $\hbar c = 197.3$ eV·nm · $hc = 1240$ eV·nm · $m_ec^2 = 0.511$ MeV · $\frac{\hbar^2}{2m_e} = 0.0381$ eV·nm² · $a = 0.0529$ nm · $\int e^{-\alpha x^2} = \sqrt{\pi/\alpha}$, $\int x^2e^{-\alpha x^2} = \frac{1}{2\alpha}\sqrt{\pi/\alpha}$ · $\int_0^\infty x^ne^{-\beta x} = \frac{n!}{\beta^{n+1}}$